Topics in Deep Learning Theory (Spring 2026)

Lecture 0: Concentration Inequalities

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Abstract

Concentration phenomena and inequalities play a pivotal role in high-dimensional probability, as they provide powerful tools for understanding the behavior of random variables and their aggregates in high-dimensional spaces. These tools allow us to bound the deviations of random variables from their expectations, ensuring that the aggregate behavior remains predictable even in complex systems. This is particularly critical in modern applications such as machine learning, statistical inference, and optimization, where datasets and models often involve thousands or millions of dimensions. Great textbooks on this topic include:

- High-Dimensional Probability by Roman Vershynin.
- Probability in High Dimension by Roman van Handel.

For more details related to the materials covered in this lecture, we specifically refer to [Vershynin, 2018, Section 2].

1 Motivation

Machine learning theory is not merely a theory of statistical estimation. Rather, it aims to understand how **data**, **model**, and **algorithm** jointly determine the final performance of a learning system.

More concretely, a typical learning pipeline consists of the following components:

- **Data.** A dataset
$$\{(x_i, y_i)\}_{i=1}^n$$
sampled from an unknown distribution.

- **Model.** A parameterized function
$$f_\theta : \mathcal{X} \to \mathcal{Y}, \qquad \theta \in \mathbb{R}^m,$$
which specifies the hypothesis class.

- **Algorithm.** A training procedure that approximately minimizes the empirical risk
$$\widehat{\mathcal{R}}_n(\theta) = \frac{1}{n}\sum_{i=1}^n \ell(f_\theta(x_i), y_i)$$
such as stochastic gradient descent (SGD):
$$\theta_{t+1} = \theta_t - \eta_t g_t, \quad t = 0, 1, \ldots, T-1,$$
where $g_t \in \mathbb{R}^m$ is a mini-batch stochastic gradient.

- **Evaluation.** The quality of the learned model is measured by the population risk

$$\mathcal{R}(\theta) = \mathbb{E}[\ell(f_\theta(X), Y)].$$

The central goal of machine learning theory is to understand how the final performance depends on key system parameters such as the sample size n, the model size m, and the number of training iterations T. In other words, we would like to understand the *scaling behavior* of learning systems.

Let $\mathcal{E}(n, m, T) = \mathcal{R}(\theta_T)$ denote the resulting prediction error. The problem can then be formulated as understanding:

How does $\mathcal{E}(n, m, T)$ scale as $n, m, T \to \infty$?

Note that the scaling behavior depends on the data structure, model architecuture, and optimization algorithms.

Understanding the scaling behavior of $\mathcal{E}(n, m, T)$ as $n, m, T \to \infty$ is a highly nontrivial problem. One reason is that the limits with respect to n, m, and T may not commute. Different orders of limits correspond to different regimes of learning.

For example:

- **Optimization limit:** fix n, m and let $T \to \infty$. This corresponds to studying the convergence of the training algorithm and whether it can reach the empirical risk minimizer.
- **Statistical limit:** fix n and let $T = \infty$, and study the behavior as $m \to \infty$. This regime investigates how the generalization error changes as the model size grows. Classical U-shaped curves and the modern double descent phenomenon address this question for certain models.
- **Joint scaling limit:** let $n, m, T \to \infty$ simultaneously according to certain scaling relations, for example $m = n^\alpha$ and $T = n^\beta$ as $n \to \infty$. This regime aims to understand the scaling laws of learning systems.

Different regimes can lead to qualitatively different behaviors of the learning dynamics and the final prediction error.

1.1 Error Decomposition

Let

$$\theta^* = \arg\min_\theta \mathcal{R}(\theta)$$

be the population risk minimizer, and

$$\hat{\theta}_n = \arg\min_\theta \widehat{\mathcal{R}}_n(\theta)$$

be the empirical risk minimizer. If θ_T is the parameter obtained after T iterations of the training algorithm, we can decompose the error as

$$f_{\theta_T} - f^* = \underbrace{f_{\theta_T} - f_{\hat{\theta}_n}}_{\text{optimization error}} + \underbrace{f_{\hat{\theta}_n} - f_{\theta^*}}_{\text{estimation error}} + \underbrace{f_{\theta^*} - f^*}_{\text{approximation error}}.$$

These three terms correspond to three different sources of error:

- **Approximation error.** This term measures how well the model class can approximate the target function. It reflects the expressive power of the model architecture.
- **Estimation error.** This term measures the discrepancy between the empirical risk minimizer and the population risk minimizer due to finite data.
- **Optimization error.** This term measures the error caused by incomplete optimization during training.

1.2 Classical and Modern Machine Learning Theory

Classical statistical learning theory primarily focuses on the *estimation error*. This perspective is largely shaped by the historical setting in which machine learning methods were developed.

- Historically, both the available data and computational resources were limited. Consequently, the learning problems that could be practically addressed were relatively simple.
- In such settings, relatively small models—such as linear predictors or kernel methods with carefully designed features—were often sufficient to approximate the target mapping with high accuracy. As a result, the approximation error was typically not a major concern.
- Moreover, because the hypothesis class was relatively simple, the associated optimization problems were usually well behaved and could be solved efficiently.

In modern machine learning, however, the situation has changed dramatically. The rapid growth of data and computational resources enables us to tackle far more complex learning tasks, such as playing Go or predicting protein folding. In these problems, the underlying mapping to be learned can be extremely complicated; for example, the policy function that maps a Go board state to the optimal next move.

To approximate such complex mappings, highly expressive neural networks are required. Consequently, the *approximation error* becomes a central issue, and the design of model architectures—such as increasing depth and width—plays a crucial role.

At the same time, the dramatic increase in model size also makes the training problem significantly more challenging. Finding parameters that approximately minimize the empirical risk is a difficult high-dimensional nonconvex optimization problem. As a result, the *optimization error* also becomes an important component of the overall learning error.

Therefore, in modern machine learning theory, both *approximation error* and *optimization error* have become central objects of study, in addition to the classical focus on estimation error.

1.3 From Estimation Error to Concentration Inequalities

In this lecture, we focus on the estimation error. A key step in analyzing estimation error is to understand the relationship between the empirical risk and the population risk. For a fixed parameter θ, this reduces to studying the deviation between an empirical average and its expectation:

$$\frac{1}{n}\sum_{i=1}^{n} X_i \quad \text{versus} \quad \mathbb{E}[X].$$

Concentration inequalities provide non-asymptotic probabilistic bounds for such deviations. They form one of the fundamental tools in statistical learning theory and high-dimensional probability.

2 Introduction

Let $X_1, \ldots, X_n$ be i.i.d. random variables with expectation μ. Then,

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mu.$$

We are interested in when *the empirical mean* $\frac{1}{n}\sum_{i=1}^{n} X_i$ *will concentrate around the population mean* μ.

- What conditions are required for the random variable X_i?
- What does the "concentration" mean?

Let first review two classical results in standard probability theory textbook.

Theorem 2.1 (Strong law of large numbers (LLN))**.** *Let $X_1, \ldots, X_n$ be a sequence of i.i.d. random variables with expectation μ. Then,*

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to \mu \quad \textit{almost surely}.$$

LLN shows that as long as the expectation μ is finite, the empirical mean will converge to μ. In other words, as long as we have sufficient samples, $\frac{1}{n}\sum_{i=1}^{n} X_i$ will always concentrate at μ. Unfortunately, the rate of "concentration" in LLN can be arbitrarily slow. The next theorem, the central limit theorem, makes one step further shows that if the second moment is finite, the deviation should be on the order of $O(1/\sqrt{n})$.

Theorem 2.2 (Central limit theorem (CLT))**.** *Let $X_1, \ldots, X_n$ be a sequence of i.i.d. random variables with mean μ and variance σ^2. Then,*

$$\sqrt{n}\left(\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right) \to \mathcal{N}(0, \sigma^2) \quad \textit{in distribution}.$$

CLT implies that $\frac{1}{n}\sum_{i=1}^{n} X_i \approx \mu + \frac{\sigma}{\sqrt{n}} Z$, where Z is the standard normal random variable. Thus, the CLT provides a precise characterization of how the empirical mean deviates from the population mean μ when the deviation is on the order of $1/\sqrt{n}$. In other words, it delivers a strong, yet asymptotic, understanding of the distribution of small deviations. However, if we are interested in large deviations—those whose magnitude does not diminish as n grows—or in non-asymptotic descriptions, the CLT alone is not sufficient.

3 Linear Concentration

Moment-based concentration. By Chebyshev's inequality,

$$\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq t\right\} = \mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right|^2 \geq t^2\right\} \leq \frac{\mathbb{E}[|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu|^2]}{t^2} \leq \frac{\sigma^2}{nt^2}.$$

This probability of having large deviations is in the order of $O(1/n)$.

However, on the other hand, from CLT, we "anticipate" that

$$\mathbb{P}\left\{|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu| \geq t\right\} \approx \mathbb{P}\left\{|\frac{\sigma Z}{\sqrt{n}}| \geq t\right\} = 2\mathbb{P}\left\{Z \geq \frac{\sqrt{n}t}{\sigma}\right\}$$
$$= \sqrt{\frac{2}{\pi}} \int_{\frac{\sqrt{n}t}{\sigma}}^{\infty} e^{-\frac{x^2}{2}}\,\mathrm{d}x \lesssim e^{-\frac{1}{2}(\frac{\sqrt{n}t}{\sigma})^2} = e^{-\frac{nt^2}{2\sigma^2}}. \tag{1}$$

This suggests that the tail can decay exponentially fast, which is much stronger than the one provided by Chebyshev's inequality. Unfortunately, this calculation is not correct since $\frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i - \mu \to \sigma Z$ can be arbitrarily slow. Therefore, we need to control somethings stronger than the second-order moments.

Let us first look at a simple example.

Theorem 3.1 (Hoeffding's inequality)**.** *Let $X_1, \ldots, X_n$ be i.i.d. symmetric Bernoulli random variable, i.e., $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{1}{2}$. Then,*

$$\mathbb{P}\left\{\frac{1}{n}\sum_{i=1}^{n} X_i \geq t\right\} \lesssim e^{-\frac{nt^2}{2}}.$$

Proof. We have

$$\mathbb{P}\left\{\frac{1}{n}\sum_{i=1}^{n} X_i \geq t\right\} = \mathbb{P}\left\{e^{\lambda\sum_{i=1}^{n}} \geq e^{n\lambda t}\right\} \leq \frac{\mathbb{E}[e^{\lambda\sum_{i=1}^{n} X_i}]}{e^{n\lambda t}}$$
$$= e^{-n\lambda t}\prod_{i=1}^{n}\mathbb{E}[e^{\lambda X_i}] = e^{-n\lambda t + n\psi(\lambda)}, \tag{2}$$

where

$$\psi(\lambda) = \log\mathbb{E}[e^{\lambda X}] = \log(\frac{e^{\lambda} + e^{-\lambda}}{2}) \leq \lambda^2/2. \tag{3}$$

Plugging it into (2), we have

$$\mathbb{P}\left\{\frac{1}{n}\sum_{i=1}^{n} X_i \geq t\right\} \leq \inf_{\lambda>0} e^{-n\lambda t + n\psi(\lambda)} = \inf_{\lambda} e^{-n(\lambda t - \lambda^2/2)} = e^{-nt^2/2}.$$

□

Remark 3.2. *The above approach is often referred as the* Chernoff-Cramer method.

From the proof, we can see that the key ingredient is the log-moment generating function (log-MGF):

$$\psi(\lambda) = \log\mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \tag{4}$$

and the Legendre conjugate of the log-MGF:

$$\psi^*(t) = \sup_{\lambda>0}\{\lambda t - \psi(\lambda)\}. \tag{5}$$

Lemma 3.3. *If X has a log-MGF ψ with the Legendre dual ψ^*, then*

$$\mathbb{P}\{X - \mathbb{E}[X] \geq t\} \leq e^{-\psi^*(t)}.$$

Let $X_1, \ldots, X_n$ be i.i.d. random variable. Then,

$$\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X]\right| \geq t\right\} \leq 2e^{-n\psi^*(t)}.$$

The above lemma implies that $\psi^*(t)$ controls the rate of concentration.

Definition 3.4 (sub-Gaussian)**.** A random variable X is said to be sub-Gaussian with variance proxy σ^2 if $\psi(\lambda) \leq \frac{\lambda^2\sigma^2}{2}$.

The sub-Gaussian assumption implies that

$$\psi^*(t) = \sup_{\lambda>0}\{\lambda t - \psi(\lambda)\} \geq \sup_{\lambda>0}\left\{\lambda t - \frac{\lambda^2\sigma^2}{2}\right\} = \frac{t^2}{2\sigma^2}.$$

By Lemma 3.3, the tail of X satisfies

$$\mathbb{P}\{|X - \mathbb{E}[X]| \geq t\} \leq 2e^{-\frac{t^2}{2\sigma^2}}, \tag{6}$$

which is similar to the tail of Gaussian. In fact, the tail estimate (6) is often used as the equivalent definition of the sub-Gaussian class.

Lemma 3.5. *If the tail behavior of X satisfies*

$$\mathbb{P}\{|X| \geq t\} \leq C_1 e^{-C_2 t^2} \text{ for all } t \geq 0. \tag{7}$$

Then, $\varphi(\lambda) \leq K_1\lambda^2$ for some constant K_1.

Proof. With loss of generality, we consider only the case of $\lambda \geq 0$. Then, we have

$$\begin{aligned}
\mathbb{E}[e^{\lambda X}] &= \int_0^\infty \mathbb{P}\left\{e^{\lambda X} \geq t\right\} \mathrm{d}t \\
&= \int_{-\infty}^\infty \mathbb{P}\left\{e^{\lambda X} \geq e^{\lambda s}\right\} \lambda e^{\lambda s}\, \mathrm{d}s \qquad (t = e^{\lambda s}) \\
&= \lambda\left(\int_{-\infty}^0 \mathbb{P}\{X \geq s\} e^{\lambda s}\, \mathrm{d}s + \int_0^\infty \mathbb{P}\{X \geq s\} e^{\lambda s}\, \mathrm{d}s\right) \\
&\leq \lambda\left(\int_{-\infty}^0 e^{\lambda s}\, \mathrm{d}s + 2\int_0^\infty e^{-C_1 s^2 + \lambda s}\, \mathrm{d}s\right) \\
&\leq 1 + C_1\lambda e^{K\lambda^2} \\
&\leq e^{K_1\lambda^2},
\end{aligned}$$

where C, K, K_1 are some absolute positive constants.

□

Corollary 3.6 (Chernoff's bound)**.** *Let $X_1, \ldots, X_n$ be i.i.d. sub-Gaussian random variables with mean μ and variance proxy σ^2. Then*

$$\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq t\right\} \leq 2e^{-\frac{nt^2}{2\sigma^2}}.$$

By Lemma 6, we can conclude that as long as each random variable has a sub-Gaussian tail, we have $P(|\frac{1}{n}\sum_i X_i - \mu| \geq t) \leq 2e^{-K_1 n t^2}$ for some constant K_1.

Examples:

- **Gaussian RV:** For $g \sim \mathcal{N}(0,1)$, its tail behavior satisfies [Vershynin, 2018, Proposition 2.1.2]
$$\left(\frac{1}{t} - \frac{1}{t^3}\right)\frac{1}{\sqrt{2\pi}}e^{-t^2/2} \le \mathbb{P}\{g \ge t\} \le \frac{1}{t}\cdot\frac{1}{\sqrt{2\pi}}e^{-t^2/2}.$$

- **Bounded RV:** Bounded random variables obviously satisfy the tail behavior (7). Specifically, the following lemma provides a tight estimate of the variance proxy.

  Lemma 3.7 (Hoeffding's lemma)**.** *Assume $a \le X \le b$. Then, $\psi(\lambda) \le \lambda^2(b-a)^2/8$.*

  Proof. WLOG, assume that $\mathbb{E}[X] = 0$. Recall that $\psi(\lambda) = \log \mathbb{E}[e^{\lambda X}]$. Then,
$$\psi'(\lambda) = \frac{\mathbb{E}[Xe^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]}, \qquad \psi''(\lambda) = \frac{\mathbb{E}[X^2e^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]} - \left(\frac{\mathbb{E}[Xe^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]}\right)^2.$$
  Let $\mathbb{Q}$ denote the distribution with $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = e^{\lambda X}/\mathbb{E}[e^{\lambda X}]$. Then, we can rewrite the second-order derivative as $\mathrm{Var}_Q[X]$. Since $X \in [a,b]$, we have
$$\mathrm{Var}_{\mathbb{Q}}[X] = \mathbb{E}_{\mathbb{Q}}[|X - \mathbb{E}_Q[X]|^2] \le \mathbb{E}_{\mathbb{Q}}\left[\left|X - \frac{b+a}{2}\right|^2\right] \le \mathbb{E}_{\mathbb{Q}}\left[\left|\frac{b-a}{2}\right|^2\right] = \frac{(b-a)^2}{4},$$
  where the second step use the fact that $\mathbb{E}[Z] = \mathrm{argmin}_\mu \mathbb{E}[(Z-\mu)^2]$ for any random variable Z. Hence,
$$\psi(0) = 0, \quad \psi'(0) = \mathbb{E}[X] = 0, \quad \psi''(\lambda) \le \frac{(b-a)^2}{4},$$
  which implies
$$\begin{aligned}\psi(\lambda) &= \psi(0) + \int_0^\lambda \psi'(t)\,\mathrm{d}t \\ &= \psi(0) + \int_0^\lambda \left(\psi'(0) + \int_0^t \psi''(s)\,\mathrm{d}s\right)\mathrm{d}t \\ &= \psi(0) + \psi'(0)\lambda + \int_0^\lambda \mathrm{d}t \int_0^s \psi''(s)\,\mathrm{d}s \le \frac{(b-a)^2\lambda^2}{8}.\end{aligned}$$
  □

Remark 3.8. *The Hoeffding's lemma is sharp when X is the symmetric Bernoulli distribution, i.e., $\mathbb{P}(X=1) = \mathbb{P}(X=-1) = 1/2$. See Eq. (3).*

Corollary 3.9 (Hoeffding's inequality)**.** *Let $X_1, \ldots, X_n$ be i.i.d. random variables. If $a \le X_i \le b$, then,*
$$\mathbb{P}\left\{\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| \ge t\right\} \le 2e^{-\frac{2nt^2}{(b-a)^2}}.$$

4 Nonlinear Concentration

Let $f : \mathbb{R}^n \mapsto \mathbb{R}$ be a (nonlinear) function and consider the following concentration:

$$f(X_1, \ldots, X_n) \approx \mathbb{E}[f(X_1, \ldots, X_n)] \quad \text{with high probability?}$$

The preceding results correspond to $f(x_1, \ldots, x_n) = \frac{1}{n}\sum_{i=1}^n x_i$. Can we extend it to nonlinear functions?

- If f only depends on one coordinate, we can not anticipate any concentration.
- If f is equally robust to small changes for all coordinates, we anticipate that this case will behave like the empirical mean.

Theorem 4.1 (McDiarmid's inequality)**.** *Let f be a function satisfying the Bounded Difference Property: there exist constants $\{L_i\}_{i=1}^n$ such that the following holds for all $i \in [n]$*

$$\sup_{x_1,\ldots,x_n,\tilde{x}_i} |f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_{i-1}, \tilde{x}_i, x_{i+1}, \ldots, x_n)| \leq L_i$$

Let $\sigma^2 := \frac{1}{4}\sum_{i=1}^n L_i^2$ and $X_1, \ldots, X_n$ are i.i.d. . Then, $Z = f(X_1, \ldots, X_n)$ is sub-Gaussian with variance proxy σ:

$$\mathbb{P}\{|Z - \mathbb{E}[Z]| \geq t\} \leq 2e^{-\frac{t^2}{2\sigma^2}}.$$

One can think L_i as a measure of the sensitivity of f to the i-th coordinates. For the case of empirical mean, $D_i f(x) = O(1/n)$ for every i. This recovers the Hoeffding's inequality (Corollary 3.9). Thus, we can viewed McDiarmid's inequality as a nonlinear version of Hoeffding's inequality. Question: Is there a similar nonlinear Chernoff's inequality?

The proof needs following lemmas.

Lemma 4.2 (Azuma's lemma)**.** *Let $\{\mathcal{F}_i\}_{i=1}^n$ be a filtration. Assume σ_i to be positive constants and $\{\Delta_i\}$ random variables such that*

1. $\mathbb{E}[\Delta_i|\mathcal{F}_{i-1}] = 0$ *(Martingale difference property).*
2. $\log \mathbb{E}[e^{\lambda\Delta_i}|\mathcal{F}_{i-1}] \leq \frac{\lambda^2\sigma_i^2}{2}$ *(Conditional sub-Gaussian property).*

Then, $\sum_{i=1}^n \Delta_i$ is sub-Gaussian with the proxy variance $\sum_{i=1}^n \sigma_i^2$.

Proof. This time, we do not have the independence. Instead, we can exploit the conditional independence, i.e., the martingale property. Consider the condition on the filtration

$$\mathbb{E}\left[e^{\lambda\sum_{i=1}^n \Delta_i}\right] = \mathbb{E}\left[\mathbb{E}[e^{\lambda\sum_{i=1}^n \Delta_i}|\mathcal{F}_{n-1}]\right]$$
$$\mathbb{E}\left[e^{\lambda\sum_{i=1}^{n-1} \Delta_i}\,\mathbb{E}[e^{\lambda\Delta_n}|\mathcal{F}_{n-1}]\right] \leq e^{\frac{\lambda^2\sigma_n^2}{2}}\,\mathbb{E}\left[e^{\lambda\sum_{i=1}^{n-1} \Delta_i}\right]$$

By induction, we conclude that

$$\mathbb{E}[e^{\lambda\sum_{i=1}^n \Delta_i}] \leq e^{\frac{\lambda^2\sum_{i=1}^n \sigma_i^2}{2}}.$$

This means $\sum_{i=1}^n \Delta_i$ is sub-Gaussian with the proxy variance $\sum_{i=1}^n \sigma_i^2$. □

Lemma 4.3 (Azuma-Hoeffding's inequality)**.** *Under the assumption of Lemma 4.2, assume $A_i \le \Delta_i \le B_i$ almost surely and A_i, B_i are $\mathcal{F}_{i-1}$-measurable. Then, $\sum_{i=1}^n \Delta_i$ is sub-Gaussian with the proxy variance $\sigma^2 = \frac{1}{4}\sum_{i=1}^n \|B_i - A_i\|_{L^\infty}$. In particular,*

$$\mathbb{P}\left\{\Big|\sum_{i=1}^n \Delta_i\Big| \ge t\right\} \le 2e^{-\frac{t^2}{2\sigma^2}}.$$

Proof. Combining Lemma 3.3, 3.7 and 4.2, we complete the proof. □

Proof of McDiarmid's inequality. To analyze the behavior of $f(X_1, \dots, X_n)$, consider the following decomposition

$$\begin{aligned} f(X) - \mathbb{E}[f(X)] &= f(X) - \mathbb{E}[f(X)|X_1, \dots, X_{n-1}] \\ &\quad + \mathbb{E}[f(X)|X_1, \dots, X_{n-1}] - \mathbb{E}[f(X)|X_1, \dots, X_{n-2}] \\ &\quad + \cdots + \mathbb{E}[f(X)|X_1] - \mathbb{E}[f(X)] \\ &= \sum_{i=1}^n \Delta_i, \end{aligned} \tag{8}$$

where $\Delta_i = \mathbb{E}[f(X)|X_1, \dots, X_i] - \mathbb{E}[f(X)|X_1, \dots, X_{i-1}]$. Let $\mathcal{F}_i = \sigma(X_1, \dots, X_i)$. Then, $\mathbb{E}[\Delta_i|\mathcal{F}_{i-1}] = 0$ and

$$\Delta_i = \mathbb{E}\Big[\mathbb{E}[f(X_1, \dots, X_i, \dots, X_n)|X_i] - f(X)|X_1, \dots, X_{i-1}\Big].$$

Let

$$\begin{aligned} A_i &= \mathbb{E}[\inf_\alpha f(X_1, \dots, X_{i-1}, \alpha, X_{i+1}, \dots, X_n) - f(X_1, \dots, X_n)|X_1, \dots, X_{i-1}] \\ B_i &= \mathbb{E}[\sup_\alpha f(X_1, \dots, X_{i-1}, \alpha, X_{i+1}, \dots, X_n) - f(X_1, \dots, X_n)|X_1, \dots, X_{i-1}] \end{aligned}$$

By the assumption of f, it is easy to verify that

$$A_i \le \Delta_i \le B_i, \qquad |B_i - A_i| \le \|D_i f\|_{L^\infty}.$$

Using the Azuma-Hoeffding lemma, $f(X) - \mathbb{E}[f(X)$ is a sub-Gaussian with the variance proxy $\sigma^2 = \frac{1}{4}\sum_{i=1}^n \|D_i f\|_{L^\infty}^2$. This directly implies that

$$\mathbb{P}\{|f(X) - \mathbb{E}[f(X)]| \ge t\} \le 2e^{-\frac{2}{\sum_{i=1}^n \|D_i f\|_{L^\infty}^2}}.$$

Thus, we complete the proof. □

5 Maximal Inequality

Lemma 5.1 (Maximal inequality)**.** *Assume that $X_1, \dots, X_n$ be n sub-Gaussian random variables with zero mean and the variance proxy σ^2. Then,*

$$\mathbb{E}[\max_{i\in[n]} X_i] \le \sigma\sqrt{2\log n}.$$

Proof. Recalling the LogSumExp trick we introduced in Lecture 3, we have for any $\beta > 0$:

$$\max_{i\in[n]} X_i \le \frac{1}{\lambda} \log \sum_{i=1}^{n} e^{\lambda X_i}.$$

For any $\lambda > 0$,

$$\begin{aligned}
\mathbb{E}[\max_{i\in[n]} X_i] &\le \frac{1}{\lambda}\, \mathbb{E}[\log \sum_{i=1}^{n} e^{\lambda X_i}] \\
&\le \frac{1}{\lambda} \log \sum_{i=1}^{n} \mathbb{E}[e^{\lambda X_i}] \qquad \text{(Jensen's inequality)} \\
&\le \frac{1}{\lambda} \log \sum_{i=1}^{n} e^{\frac{\sigma^2\lambda^2}{2}} = \frac{\log n}{\lambda} + \frac{\sigma^2\lambda}{2}.
\end{aligned}$$

Taking $\lambda = \sqrt{2\log(n)/\sigma^2}$ completes the proof. □

Note that in the maximal inequality, we do not assume that $X_1, \ldots, X_n$ are independent. In fact, the bound in Lemma 5.1 is sharp.

Lemma 5.2. *Let $X_1, \ldots, X_n$ be independent $\mathcal{N}(0,1)$ random variables. Then,*

$$\mathbb{E} \max_{i\in[n]} X_i \ge c\sqrt{\log n}.$$

Next, we turn to provide high-probability bound.

Lemma 5.3. *Let $X_1, \ldots, X_n$ be sub-Gaussian random variables with zero mean and variance proxy σ^2. Then, for any $\delta \in (0,1)$, with probability at least $1-\delta$, it holds that*

$$\max_{i\in[n]} X_i \le \sigma\sqrt{2\log(\frac{2n}{\delta})}.$$

Proof. By union bound, we have

$$\mathbb{P}\{\max_i X_i \ge t\} \le \sum_{i=1}^{n} \mathbb{P}\{|X_i| \ge t\} \le 2ne^{-t/(2\sigma^2)}.$$

For this probability to be smaller than δ, we quire $2ne^{-t/(2\sigma^2)} \le \delta$. It gives $t \ge \sqrt{2\sigma^2\log(2n/\delta)}$. □

References

[Vershynin, 2018] Vershynin, R. (2018). *High-dimensional probability: An introduction with applications in data science*, volume 47. Cambridge university press.